

Intro to Analysis of Variance (ANOVA)

The study

PREVEND cohort • 500 Dutch adults

RFFT: connect dots into unique designs



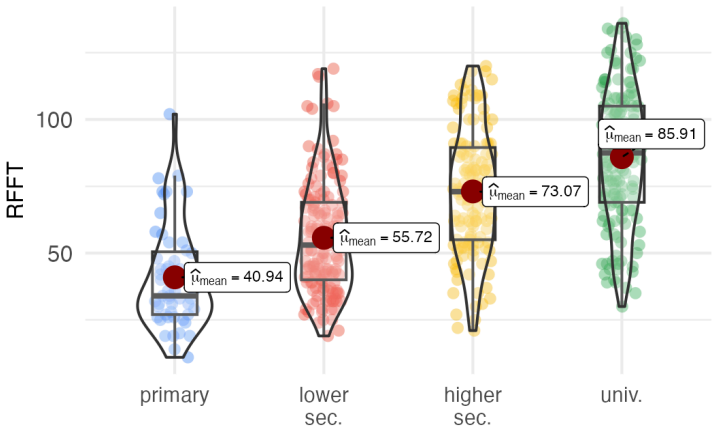
cognition score
more designs =
higher score

```
pacman::p_load(tidyverse, ggstatsplot)
pacman::p_load_gh("the-graph-courses/rsr")

rsr::prevend |>
  group_by(education) |>
  summarise(n = n(), mean = mean(rfft), sd = sd(rfft))

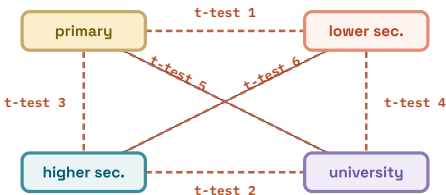
#> education      n mean  sd
#> primary         51 40.9 19.4
#> lower_secondary 157 55.7 20.9
#> higher_secondary 134 73.1 23.6
#> university      158 85.9 25.0
```

```
ggbetweenstats(data = rsr::prevend, x = education, y = rfft,
  results.subtitle = F, pairwise.display = "none")
```



Does mean cognition differ across the four education groups?

Pairwise t-tests and false-alarm risk



$P(\geq 1 \text{ false positive}) = 1 - 0.95^6 \approx 26\%$

ANOVA hypotheses

One-way ANOVA (analysis of variance) tests whether **k group means** are all equal, with one F test instead of many pairwise t-tests.

NULL HYPOTHESIS

$H_0 : \mu_{\text{prim}} = \mu_{\text{lsec}} = \mu_{\text{hsec}} = \mu_{\text{univ}}$

ALTERNATIVE HYPOTHESIS

$H_1 : \text{at least one } \mu \text{ differs}$

The F statistic | A Large F is evidence of true difference

F RATIO

$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$

MEAN-SQUARE FORMULAS

$MS_{\text{between}} = \frac{\sum n_j (\bar{y}_j - \bar{y})^2}{k - 1}$
 $MS_{\text{within}} = \frac{\sum \sum (y_{ij} - \bar{y}_j)^2}{N - k}$

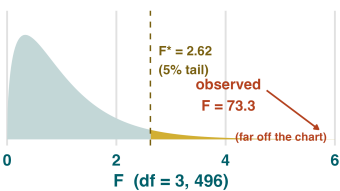
IN PLAIN LANGUAGE

MS_{between} =
group means from grand mean
group df
 MS_{within} =
people from own group mean
within df

PREVEND CALCULATION

$MS_B = 115,041/3 = 38,347$
 $MS_W = 259,469/496 = 523$
 $F = 38,347/523 = 73.3$

F curve



HOW LARGE IS LARGE?

around 1 between and within variation look similar
> 2.62 significant at $\alpha = .05$ for $df = (3, 496)$
73.3 our PREVEND sample

How to do it in R

```
fit <- aov(rfft ~ education, data = rsr::prevend)
summary(fit)
#>               Df Sum Sq Mean Sq F value Pr(>F)
#> education      3 115041   38347    73.3 <2e-16 ***
#> Residuals     496 259469     523
```

Mean RFFT scores differed significantly across education groups, $F(3, 496) = 73.3$, $p < .001$. The test tells us at least one true mean differs.

Practice: read an ANOVA

```
PlantGrowth |>
  group_by(group) |>
  summarise(n = n(), mean = mean(weight))

#> group      n mean
#> ctrl      10  5.03
#> trt1       9  4.66
#> trt2      10  5.53

summary(aov(weight ~ group, data = PlantGrowth))
#>               Df Sum Sq Mean Sq F value Pr(>F)
#> group          2   3.766   1.8832   4.846 0.0159 *
#> Residuals     27 10.492   0.3886
```

1. How many groups are being compared?
2. What are the between and within df?
3. Is the omnibus test significant?
4. Write one sentence reporting the result.

Solutions: 3 groups; df = 2 and 27; yes, $p = .0159$; plant weights differed by treatment group, $F(2, 27) = 4.85$, $p = .016$.

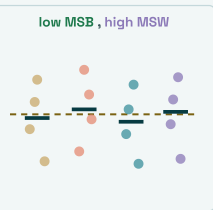
Tukey follow-up

ANOVA tells us that at least one education group differs. Tukey's HSD then checks which pairs differ, while adjusting for the fact that we are making several comparisons. Here, all six pairs differ.

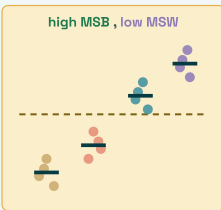
```
TukeyHSD(fit)
#>               diff      lwr      upr    p adj
#> lower_secondary-primary 14.78   5.28  24.28  4e-04
#> higher_secondary-primary 32.13  22.43  41.83  0.0000
#> university-primary      44.96  35.47  54.46  0.0000
#> higher_secondary-lower_secondary 17.35  10.42  24.29  0.0000
#> university-lower_secondary 30.19  23.54  36.83  0.0000
#> university-higher_secondary 12.83   5.91  19.75  1.4e-05
```

WHAT WOULD F LOOK LIKE?

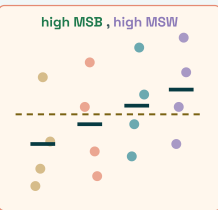
low F?



medium F?



high F?



$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$